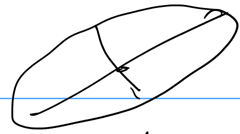
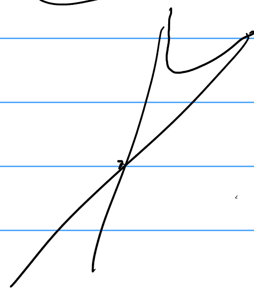
H^2 $\quad \mathbb{R}^2 \circ \{$elipsy: vol $= 1$ / środek w $0\}$



$AdS^2 \circ \{$pary prostych przez $0\}$

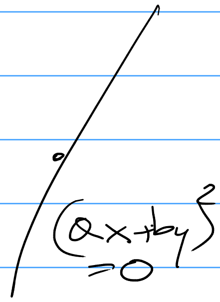
hiperbole o środku w 0



Anti de Sitter $\circ$ Lorentz

$RP^2 \circ$ "podwójne proste" przez 0

$$\left| \begin{array}{c} Q(x,y)=1 \\ x^2+y=1 \end{array} \right| : \begin{array}{c} I(x,y)=0 \\ (ax+by)(cx+dy)=0 \end{array}$$

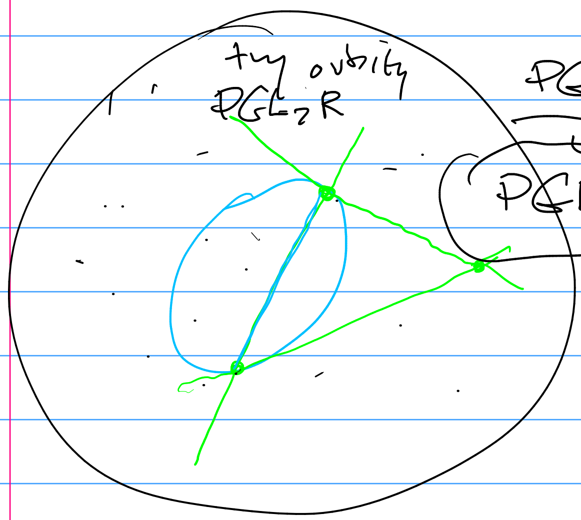


$$Q \leftrightarrow \begin{pmatrix} a & b \\ b & c \end{pmatrix} - 0/\sim \; : \; \frac{RP^2}{Gl_2(R)}$$

$$\left| \begin{pmatrix} \alpha & \beta \\ \gamma & -\alpha \end{pmatrix} \right. T\, PGL_2R$$

$\downarrow$ det

$-\alpha^2 - \gamma\beta \; (--+)$



$PGL_2 Q$

$\cup$

$PGL_2 \mathbb{Z}$

$\circ \longleftrightarrow$ proste w H^2

$\mathbb{R}P^2$

$\cup$

$QP^2 =$ całkowitoliczbowe formy

$(\mathbb{Z}^3 - 0) / \begin{pmatrix} a & b \\ b & c \end{pmatrix}$

$\det \begin{pmatrix} a & b \\ b & c \end{pmatrix}$ · zachowywane przez $PGL^{\pm}_2 R$

$\cup$

$PGL_2 \mathbb{Z}$

$a, b, c \in \mathbb{Z}$

$\gcd(a,b,c) = 1$ $\quad | \; GL_2 \mathbb{Z} \; | \; SL_2 \mathbb{Z}$

= kraty w $\mathbb{C}$
z dokładnością do działan $\mathbb{C}^*$

$x^2+y^2=0$ ~ $\pm i$ | $ax^2+bxy+cy^2=0$

$(x+iy)(x-iy)=0$ | $(x^2+y^2)(\pm i, 1)=0$

$Q(x,y)[i,1]=0 \iff Q(x,y)=a \cdot$ okrąg

$\{\text{ellipsa}\} \mapsto \pm z \ (\operatorname{Im} z \neq 0)$

$(1, z)$ · kraty · $SL_2\mathbb{Z}$ działa

$SL_2 \circlearrowleft \mathbb{Z}^2 \to \mathbb{C}$

$(1,0) \mapsto 1$

$(0,1) \mapsto z$

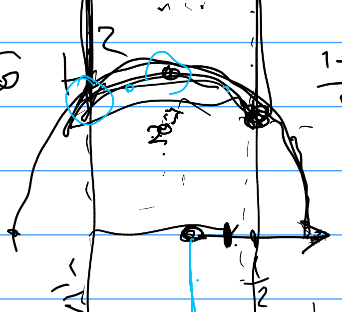
$x+k$

$PSL_2\mathbb{Z} \circlearrowleft$

$\frac{1+\sqrt{3}}{2}$

$\left(\frac{1}{z}\right)$

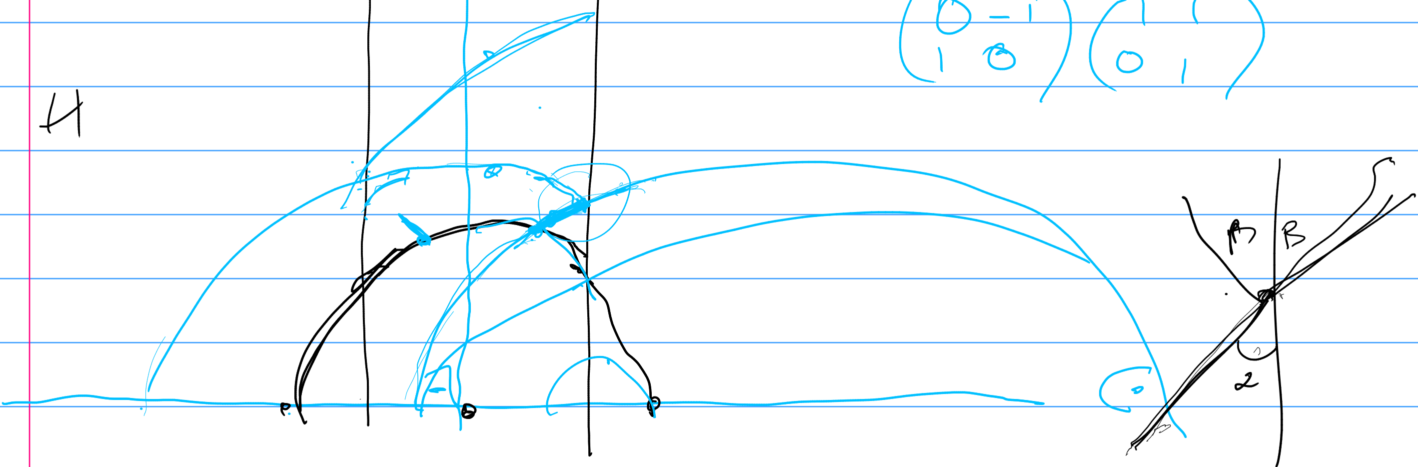
$\begin{matrix} 0 & 1 \\ -1 & 0 \end{matrix}$



$PSL_2\mathbb{Z}$

$PGL_2\mathbb{Z}$

$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$

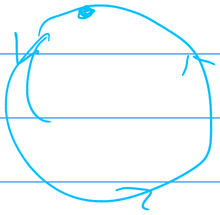


cykl form indef

indef. quadratics ∘ closed geodesics

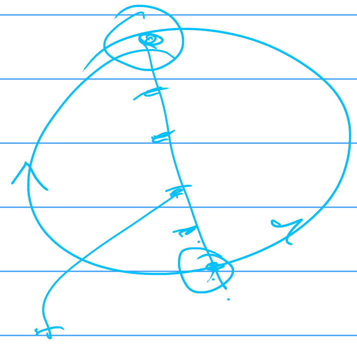
w $H^2/PSL_2(\mathbb{Z})$

$PSL_2(\mathbb{Z}) \circ \gamma$ hiperboliczny

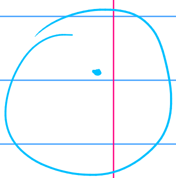


↓

indef.



H^2/PSL_2 zamknięta geodezyjne!



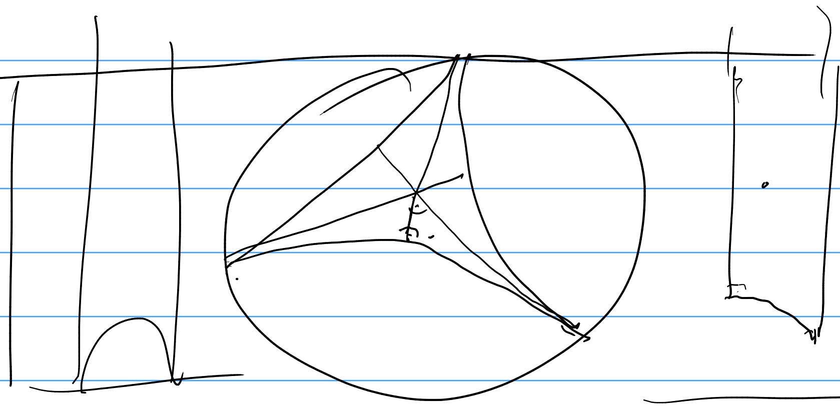
ell

++ H^2 +− AdS

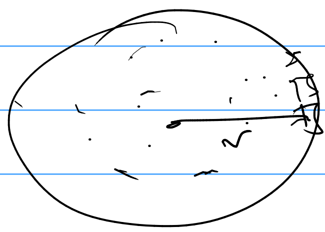
$\mathbb{Q}$ $PSL_2\mathbb{Z}$ $PSL_2\mathbb{Z}$ kiepskie

całkowitoliczbowo.

$SL_2 \ni \gamma \mapsto$ oś =



$\mathbb{R}^2 \supset \mathbb{Q}^2 \supset \mathbb{Z}^2$



$\# \mathbb{Z}^2 \cap \mathcal{O}_r$

$\approx \pi r^2$

$+ O(r)$

poprawa!